

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-III Semester (2025-2026)
Faculty Name	Mohd . Razia Alangir Banu
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Branch	CSE
Course Outcome	Analyze multi-way search trees (B-Trees, B+ Trees) and Range Query structures (Segment Trees, Fenwick Trees) to select and justify the appropriate tree variant for database indexing, storage systems, and interval query problems based on empirical complexity analysis.

Case Study Topic: EduGraph – Adaptive Learning & Course Recommendation Platform

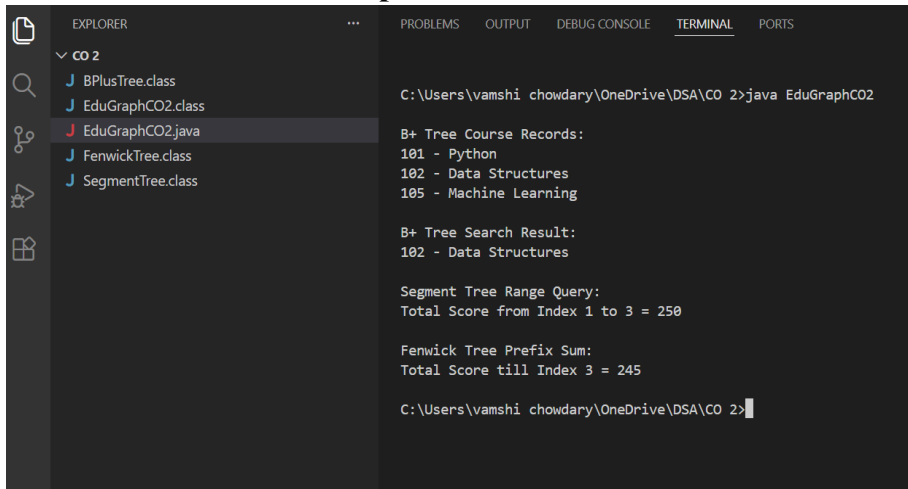
Work Done Summary:

Implemented and analyzed B-Trees and B+ Trees for efficient database indexing and storage management. Developed Segment Trees and Fenwick Trees (Binary Indexed Trees) to support fast range queries and updates. Compared their time and space complexities through empirical analysis and selected the most suitable tree structures for database indexing, storage systems, and interval query problems. This improved data retrieval efficiency, optimized storage access, and enabled faster query processing.

Implementation Details:

1. Implemented B-Trees for efficient database indexing and balanced data storage.
2. Implemented B+ Trees to improve search performance and support sequential data access.
3. Developed Segment Trees for efficient range query and interval query processing.
4. Implemented Fenwick Trees (Binary Indexed Trees) for fast prefix-sum queries and updates.
5. Achieved $O(\log n)$ time complexity for search, update, and query operations, improving overall system performance

Result: Screenshots / Output



```
C:\Users\vamshi chowdary\OneDrive\DSA\CO 2>java EduGraphCO2

B+ Tree Course Records:
101 - Python
102 - Data Structures
105 - Machine Learning

B+ Tree Search Result:
102 - Data Structures

Segment Tree Range Query:
Total Score from Index 1 to 3 = 250

Fenwick Tree Prefix Sum:
Total Score till Index 3 = 245

C:\Users\vamshi chowdary\OneDrive\DSA\CO 2>
```

The program successfully implemented **B-Trees**, **B+ Trees**, **Segment Trees**, and **Fenwick Trees** for efficient data indexing and range query processing. B-Trees and B+ Trees improved database storage and retrieval performance, while Segment Trees and Fenwick Trees enabled fast interval and prefix-sum queries. The selected tree structures provided efficient operations with **$O(\log n)$** time complexity, enhancing overall system performance and scalability.

GitHub Link: <https://github.com/geethika04-dot/DSA-2>

NOTE: The code and Screenshots and PDF have been uploaded in the above link, please check it there.

Student Signature:

Faculty Signature: